Open Science in Practice: Collaborative Research with Git



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AI assistance

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Human assistance from Robert Svebeck

Quiz: AI words

- AI: Computers doing **human-like** tasks
- Algorithms: Step by step **recipe**
- Weights: **Constants** in a function, created from learning
- Model: **Architecture** plus **weights**, wrapped in **code**
- Deep Learning: Machine learning using **networks inspired by the brain.**
- Generative AI: Deep learning model that creates **new content**
- Agentic AI: Generative AI that perform tasks **on its own**
- AI Swarms: **Multiple** AI agents **coordinating** with each other

Using generative AI in your research project

- Use a chatbot as your “Sounding board”
- Convert *a “mess”* of information to structured information
- Harmonise different structured content
- Convert file formats (eg. excel -> json)
- Roleplay - Prepare for difficult conversations (reviewer, opponent)
- Generate "what if" scenarios
- Validate a text against checklists
- Create checklists and templates

“… every time I delegated the writing (to the AI) I missed out on much of the thinking that I would have done while trying to think through or explain an idea.” / Anonymous Researcher

Regarding hallucinations

Hallucinations is a *bi-product* not a *bias*

- Generative AI produces likely outputs based on patterns learned from training data and the prompt
- Not necessarily fact checking



Who was the sole survivor of Titanic?

The sole survivor of the Titanic disaster was Charles Joughin. He was a British baker and the chief baker aboard the Titanic. Joughin managed to survive the sinking by reportedly clinging to an overturned lifeboat and enduring several hours in the frigid water before being rescued by the RMS Carpathia. His survival is often attributed to his high alcohol consumption, which may have helped him withstand the extreme cold. Joughin lived until 1956 and later provided testimonies about his experiences during the Titanic disaster.

You

who was the sole survivor of Titanic?

ChatGPT

The term "sole survivor" of the Titanic is often used to refer to Millvina Dean, who was the last living survivor of the Titanic disaster. Millvina Dean was born on February 2, 1912, and she was only 2 months old when the Titanic sank in April 1912. She, along with her mother and brother, were among the survivors.

Millvina Dean passed away on May 31, 2009, at the age of 97, making her the last remaining survivor of the Titanic. Her story, along with the tragic events of the Titanic, has been the subject of much historical interest and fascination.

Regarding reasoning errors

I need to wash my car. It is only 400 meters to the car wash. Should I drive there or walk?

Mistral Small 4

Since the car wash is only 400 meters away, walking would be a reasonable and eco-friendly choice. It's a short distance that won't take much time or effort, and it avoids unnecessary fuel consumption. If you have time and good weather, walking could be a nice way to get some light exercise as well.

2nd opinion

I need to wash my car. It is only 400 meters to the car wash. Should I drive there or walk?

Claude Haiku 4.5

Whether you should drive or walk depends on a few factors worth considering:

Drive if:

- You're short on time or in a hurry
- It's bad weather (rain, extreme heat, cold)
- You're carrying supplies or need to transport items
- You're physically tired or have mobility concerns
- The car wash is in an unsafe area to walk

Walk if:

- You have time to spare — it's only about a 5-minute walk at a normal pace
- The weather is pleasant
- You want the exercise and fresh air
- You're trying to reduce your carbon footprint
- Parking at the car wash might be inconvenient

The practical case for walking: At 400 meters, walking is quite reasonable. You'd spend more time driving (starting the car, backing out, driving, parking) than simply walking there. Plus, you'll save fuel and parking hassle.

I need to wash my car. It is only 400 meters to the car wash. Should I drive there or walk?

GPT-5.6 Luna

Drive there—the car needs to be at the car wash.

2nd opinion

Processing sensitive data with GenAI



If you ask a *friend* for advice...

Research Institute

Researcher by his desk, with a unpublished document

Needs a second opinion from an expert.

→

“External Expert Office”

Shows the document to the expert

The expert reads it and gives an opinion.

→

Research Institute

Back at the desk, document in hand

It was never left behind. It is "stored locally".

Question: Was there a data leak?

What the researcher could not see

A photocopier

A copy takes seconds. He would never know.

Cloud Compute:
logs, caches, prompt retention

Cameras in the room

Someone else may watch while he shows it.

Cloud Compute:
monitoring, telemetry, abuse checks

Another country's law

Local police can demand copies, and forbid anyone to tell.

Cloud Compute:
US CLOUD Act, FISA 702

Storage is where data rests.
-> Compute is where data is **exposed**.

Agentic AI

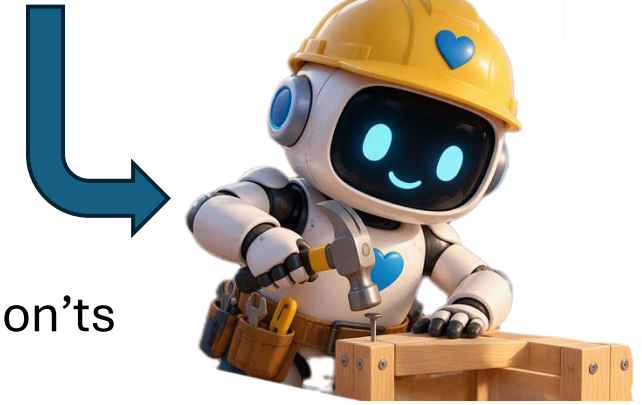
AI Agents - *in general*

- An agent is a computer program that has access to a LLM and also has access to “digital tools”
  - A tool can be a mail system, a browser, screen capture, the mouse, a microhpone, the speaker, a EHR system, a database, your holiday house heater system, etc
- An agent always has a starting goal
  - The goal is expressed as a LLM prompt
  - The goal does not have to be created by a human
- An agent uses the LLM to understand *unstructured* context, and then it can use this understanding to “decide” what to do with a given task
- An agent can be given a tool that will update it’s own goal

AI Agents – basic features







Every agent has a prompt

- Specific instructions, goals, dos and don’ts
- Can delegate to other agents

Still LLM so same risk of hallucinations and reasoning errors

AI Agents – basic features



“You are the orchestrator. You receive instructions, set up a plan and delegate to other agents.”



“You reason and break down difficult tasks.”



“You audit and review changes, give a second opinion.”



“You edit code per instruction from orchestrator.“

Coding examples

- Use it as **coding assistant**
- **Explain** code for you
- Create **test** scenarios
- **Debug** and explain errors
- **Convert** between formats or code-languages
- Generate **training** materials
- Use it to **generate synthetic** data

Code of conduct [draft]

- Stand behind every idea and line of code.
- Understand the code; document it and learn how it works.
- Never give AI tools access to sensitive data or information.
- Test and review the code.
- Be transparent about AI's role.

"explain this"

R: testthat; Python: pytest

eg. AGENTS.md

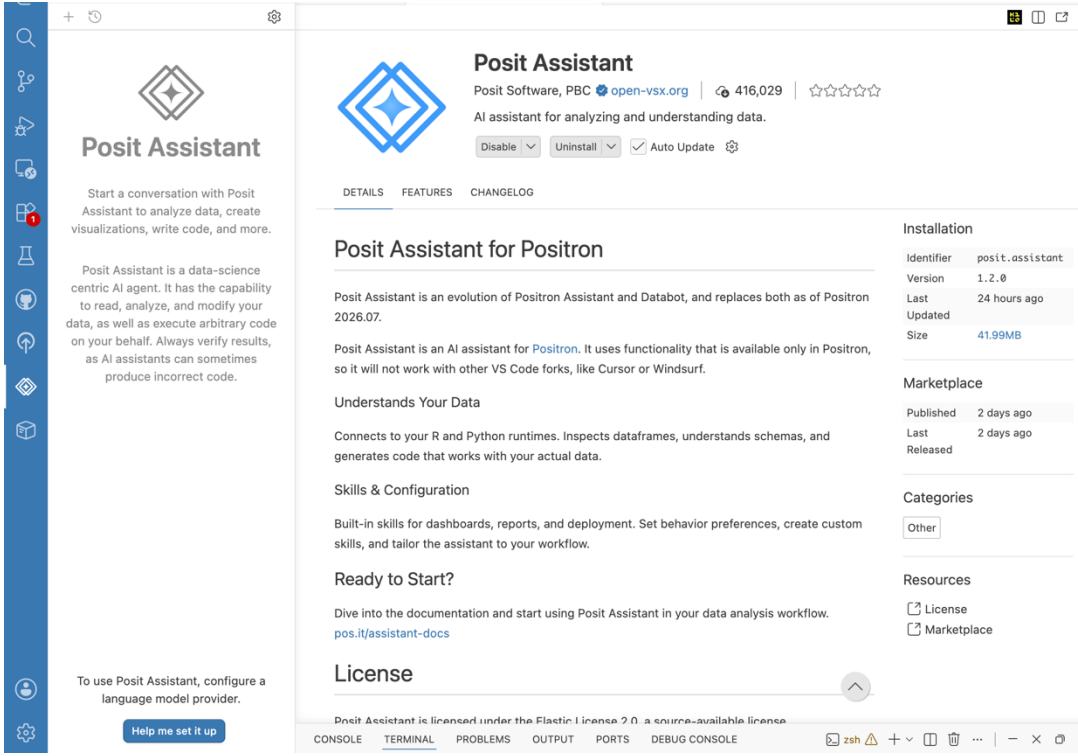
AI and Git (a few benefits)

Change tracking: Commits record what an agent changed, making its work inspectable and attributable.

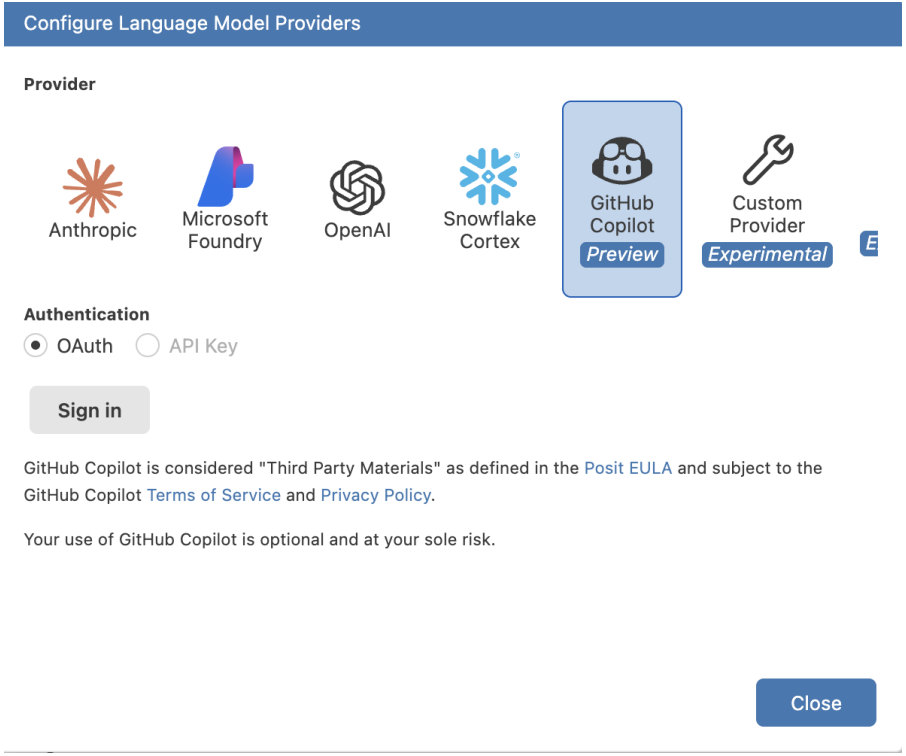
Save resource: Read changes and diffs instead of re-reading the full repository

Recovery and safety: Reverts and history make it easier to undo faulty changes and restore known-good versions.

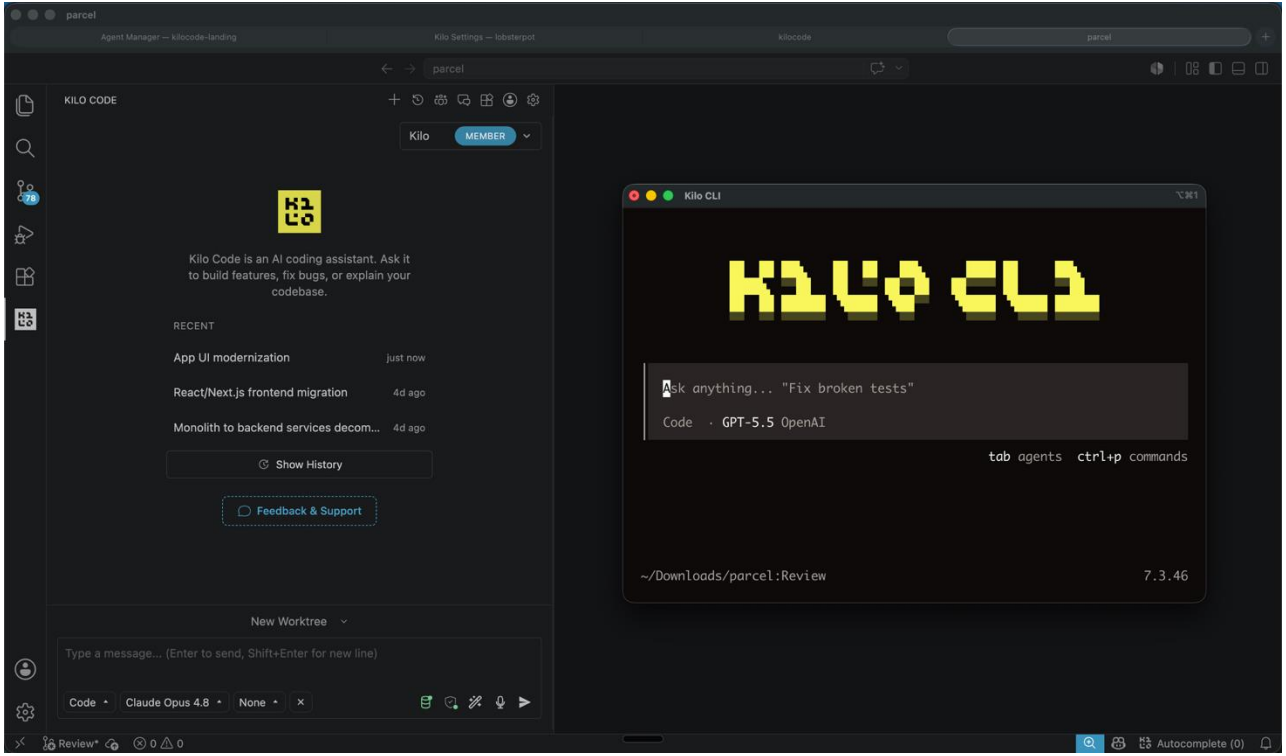
Code assistance in Positron



Code assistance in Positron



Code assistance in Positron



Want to learn more about Agentic AI?

Open Science in Practice: Responsible Use of Agentic AI in Research

https://doctoralcourses.application.ki.se/fubasextern/info?kurs=K8F6133

Centre for AI innovation

https://ki.se/en/research/research-areas-centres-and-networks/research-centres/centre-for-ai-innovation

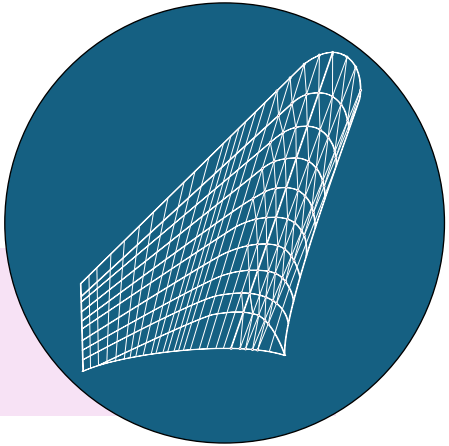
Centre for AI Innovation Karolinska Institutet

Support for faster development and use of AI in science and healthcare

Service for acadamy, healthcare and industry



Courses and training
for employees and students



Support AI innovation



Collaboration and matchmaking



Testing AI solutions



Health data services



Expertise and guidance